FLATNESS OF A STANDARD ÉTALE ALGEBRA IN TWO VARIABLES

Abstract. Let A be any commutative ring, $F, G \in A[X,Y]$ and $J = F_X G_Y - F_Y G_X$. We show that $A[X,Y][1/J]/(F,G)$ is a flat A-module. The proof is elementary and constructive: flatness is the existence of explanations of finite relations; the polynomial ring and its localization are flat; a relation in the quotient is corrected modulo (F,G) using the syzygies of a regular pair; and the regularity of F, G modulo every finitely generated ideal of A comes from finite Taylor lifts of the curve $F = f - T$, $G = g$ and a noetherian argument in a finitely generated subring. No hypothesis is made on A. The argument is due to Claude Quitté. Everything is checked in Unit.

1. Statement

Throughout, A is a commutative ring, $R = A[X,Y]$, $F, G \in R$, $J = F_X G_Y - F_Y G_X$ (formal partial derivatives), $S = R[1/J]$, and

$$B = S/(F,G).$$

We write f, g for the images of F, G in S, and u for the inverse of J in S. No condition is put on J: when $J = 0$, S and B are the zero ring.

Definition 1.1. Let C be an A-algebra. A *relation* in C is a family $a_1, \dots, a_n \in A$, $b_1, \dots, b_n \in C$ with $\sum_i a_i b_i = 0$. An *explanation* of it is a family $c_{ij} \in A$, $y_j \in C$ $(1 \le j \le m)$ with

$$b_i = \sum_j c_{ij} y_j \quad (\text{all } i), \qquad \sum_i a_i c_{ij} = 0 \quad (\text{all } j).$$

C is *flat* over A when every relation merely has an explanation.

This is the usual equational criterion, and it is equivalent to the flatness of the A-module underlying C in the form used for modules (one relation explained by a factorisation through a free module).

Theorem 1.2. *For every commutative ring A and all $F, G \in A[X,Y]$, the algebra $B = A[X,Y][1/J]/(F,G)$ is flat over A.*

No noetherian hypothesis, decidable equality, or condition on the characteristic is used.

2. Explanations, polynomials, localization

Lemma 2.1. *Fix $a_1, \dots, a_n \in A$. The families (b_i) for which the relation is explained contain 0 and are closed under sums, under multiplication by an element of C, and under finite sums. If $\varphi : C \to D$ is a map of A-algebras, the image of an explained family is explained.*

Proof. For a sum, concatenate the two lists of pairs $(c_{\cdot j}, y_j)$; for $t \cdot b_i$, replace y_j by $t y_j$; for an image, replace y_j by $\varphi(y_j)$. □

Lemma 2.2. *If C is flat over A, so is $C[X]$. Hence $A[X,Y]$ is flat over A.*

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Proof. Let $\sum_i a_i g_i = 0$ in $C[X]$. Each g_i is merely the class of a list of coefficients, so there is merely an L such that all coefficients of all g_i vanish from degree L on. Induction on L. If $L = 0$ all g_i are 0. Otherwise write $g_i = c_i + Xt_i$, with c_i the constant term. The constant term of the relation gives $\sum_i a_i c_i = 0$ in C, and the coefficients of X^{k+1} give $\sum_i a_i t_i = 0$ in $C[X]$, where the coefficients of the t_i vanish from degree $L-1$ on. So (c_i) is explained in C, hence in $C[X]$, and (t_i) by induction; by Lemma 2.1, so is $(c_i + Xt_i)$. Start from A, which is flat over itself (one pair $(b_i, 1)$). □

Lemma 2.3. *If C is flat over A and $j \in C$, then $C[1/j]$ is flat over A.*

Proof. Let $\sum_i a_i b_i = 0$ in $C[1/j]$. Finitely many fractions merely have a common denominator: $b_i = p_i/j^d$. Then $\sum_i a_i p_i/j^d = 0$, so $j^e \sum_i a_i p_i = 0$ in C for some e. The relation $\sum_i a_i (j^e p_i) = 0$ in C is explained; its image in $C[1/j]$, multiplied by $1/j^{d+e}$, explains (b_i) (Lemma 2.1). The element j need not be regular. □

In particular $S = A[X, Y][1/J]$ is flat over A.

3. Correction of a relation

Lemma 3.1 (Syzygies modulo an ideal)**.** *Let I be an ideal of a ring C and $f, g \in C$ with: $fs \in I$ implies $s \in I$, and $gv \in (f) + I$ implies $v \in (f) + I$. If $fu + gv \in I$, then merely there is w with $u - gw \in I$ and $v + fw \in I$.*

Proof. $gv = (fu + gv) - fu \in (f) + I$, so $v - ft \in I$ for some t. Then $f(u + gt) = (fu + gv) - g(v - ft) \in I$, so $u + gt \in I$. Take $w = -t$. □

Proposition 3.2. *Let C be a flat A-algebra and $f, g \in C$ such that for every finite family $a_1, \ldots, a_m$ in A, with $I = (a_1, \ldots, a_m)C$, f is regular modulo I and g is regular modulo $(f) + I$ in the sense of Lemma 3.1. Then $C/(f, g)$ is flat over A.*

Proof. Let $\sum_i a_i b_i = 0$ in $C/(f, g)$ and lift b_i to $p_i \in C$. Then $\sum_i a_i p_i = fu + gv$ for some u, v. With $I = (a_1, \ldots, a_n)C$, $fu + gv \in I$, and Lemma 3.1 gives w with

$$u - gw = \sum_i a_i u_i, \qquad v + fw = \sum_i a_i v_i.$$

Put $p_i' = p_i - fu_i - gv_i$. Each p_i' still lifts b_i, and

$$\sum_i a_i p_i' = (fu + gv) - f(u - gw) - g(v + fw) = 0 \quad \text{in } C.$$

This relation is explained in C; its image explains (b_i). □

So Theorem 1.2 follows from Lemmas 2.2 and 2.3 and the following proposition.

Proposition 3.3. *For every commutative ring A and $a_1, \ldots, a_m \in A$, with $I = (a_1, \ldots, a_m)S$: f is regular modulo I, and g is regular modulo $(f) + I$.*

4. Finite Taylor lifts

Let E be an A-algebra and $x_0, y_0 \in E$ with $J(x_0, y_0)$ invertible; put $h = F(x_0, y_0)$ and $k = G(x_0, y_0)$.

Lemma 4.1. *For every N there are $x(T), y(T) \in E[T]$ with $x(0) = x_0$, $y(0) = y_0$ and*

$$F(x(T), y(T)) \equiv h - T, \qquad G(x(T), y(T)) \equiv k \pmod{T^{N+1}}.$$

Moreover the lift of order $N+1$ is obtained from the lift of order N by adding aT^{N+1} to x and bT^{N+1} to y; so the lifts are compatible modulo T^{N+1}.

Proof. Induction on N, starting with the constants x_0, y_0. At order $N+1$, changing x, y by aT^{N+1}, bT^{N+1} changes the coefficients of T^{N+1} in $F(x,y)$ and $G(x,y)$ by $F_X a + F_Y b$ and $G_X a + G_Y b$ evaluated at (x_0, y_0), and no lower coefficient. The 2×2 system for the two errors has determinant $J(x_0, y_0)$, a unit. □

Lemma 4.2 (Kill)**.** *Let $(f - T)p(T) \equiv 0 \pmod{T^{N+1}}$ in $E[T]$, where $f = F(x_0, y_0)$, and w the coefficient of T^N in p. Then $p(0) = f^N w$ and $f^{N+1} w = 0$. If, for a sequence of such data $w_0, w_1, \ldots$ (one for each N, with the same $p(0)$), some w_k is a linear combination of $w_0, \ldots, w_{k-1}$, then $p(0) = 0$.*

Proof. Comparing coefficients, $p_n = f p_{n+1}$ for $n < N$ and $f p_0 = 0$; this gives the two equations. If $w_k = \sum_{j<k} \lambda_j w_j$, then $f^k w_j = f^{k-1-j} f^{j+1} w_j = 0$ for $j < k$, so $p(0) = f^k w_k = 0$. □

A sequence has such a good initial segment in a noetherian ring in the constructive sense of Coquand and Persson (the empty list is barred by the good lists). This applies inside a subring generated by finitely many elements, which is a quotient of $\mathbb{Z}[t_1, \ldots, t_r]$ and is noetherian in this sense by Hilbert's basis theorem, provided each w_N is merely given as a polynomial in the generators.

5. Regularity

Lemma 5.1 (Kernel principle)**.** *Let $\varphi : R \to E$ be a ring map with $\varphi(J)$ invertible and $\varphi(a_i) = 0$, and put $x_0 = \varphi(X)$, $y_0 = \varphi(Y)$.*

1. *If $FP = \sum_i a_i Q_i$ in R, then $\varphi(P) = 0$.*
2. *If $\varphi(F) = 0$ and $GP = FW + \sum_i a_i Q_i$ in R, then $\varphi(P) = 0$.*

Proof. (1) Let $E' \subseteq E$ be the subring generated by x_0, y_0, the inverse of $J(x_0, y_0)$, and the images of the coefficients of F, G, P. It is finitely generated, every element of E' is merely a polynomial in the generators, and the inclusion $E' \to E$ is injective. Run the lifts of Lemma 4.1 in E', for F, G with coefficients in E'. For any $x(T), y(T) \in E[T]$, the map $R \to E[T]$, $X \mapsto x(T)$, $Y \mapsto y(T)$, sends each a_i to 0, so $F(x,y)P(x,y) = 0$ in $E[T]$; since $E' \to E$ is injective, the same holds in $E'[T]$ for the lifts. Thus $(f - T)P(x(T), y(T)) \equiv 0 \pmod{T^{N+1}}$ for every N, with $P(x(T), y(T))$ of constant term $P(x_0, y_0) = \varphi(P)$, and Lemma 4.2 gives $\varphi(P) = 0$ in E', hence in E.

(2) The same, with the roles of F and G exchanged (Jacobian $-J$): since $F(x_0, y_0) = 0$, the lift of Lemma 4.1 gives $G \equiv g - T$ and $F \equiv 0$, so the identity becomes $(g - T)P(x,y) \equiv F(x,y)W(x,y) \equiv 0$ modulo T^{N+1}. □

Proof of Proposition 3.3. f modulo I. Let $fs \in I$. Write $s = p/J^d$ with $p \in R$; then $J^d f s$, the image of Fp, lies in I. An element of R whose image lies in I satisfies $J^k r = \sum_i a_i Q_i$ in R for some k, Q_i: write the coefficients with a common denominator and clear it with the kernel of the localization. So $F(J^k p) = \sum_i a_i Q_i$. Apply Lemma 5.1(1) to $E = S/I$: $J^k p = 0$ in S/I, so $p = 0$ there, and $s \in I$.

g modulo $(f) + I$. Let $gv - ft \in I$ and $v = p/J^d$. Then the image of Gp lies in $(f) + I$, so $J^k G p = FW + \sum_i a_i Q_i$ in R. Apply Lemma 5.1(2) to $E = S/(f) + I$ (where F vanishes): $J^k p = 0$, so $p = fc + \sum_i a_i c_i$ in S, and $v - f(u^d c) = u^d(p - fc) \in I$. □

This completes the proof of Theorem 1.2. □

Remark 5.2. The argument also gives, for every finitely generated ideal I of A, $IS \cap (F,G)S = F\,IS + G\,IS$.

6. Formalisation

Everything above is checked in Unit, in `unit/sag/etale-flat`, over univalence and replacement (the parameters common to all `sag/` developments), with no other hypothesis:

Theorem 1.2	`EtaleFlat2.Over.flat`, `EtaleFlat2.Over.module-flat`
equivalence of the two forms of flatness	`FlatRelations.Over.equivalent`
Lemma 2.1	`FlatRelations` (`Ops`, `Image`)
Lemma 2.2	`PolynomialFlat`
Lemma 2.3	`LocalizationFlat`
Lemma 3.1	`PairSyzygy`
Proposition 3.2	`QuotientFlat`
Lemma 4.1	`TaylorTargets`, `TaylorCompatibility`
Lemma 4.2	`StableAnnihilator.Kernel`, `CodedSubring.Vanish`
Lemma 5.1	`RegularPolynomial`, `RegularCoded`
Proposition 3.3	`RegularPair`, `LocalizedReduction`

Hilbert's basis theorem in the sense of Coquand–Persson is `sag/noetherian`; the coded subring and Lemma 4.2 along a merely coded sequence are `sag/jacobian/CodedSubring`, following `doc/jacobian-noetherian.pdf`. Equalities are equalities in the actual rings (coefficient quotients, fractions, quotient rings), not of polynomial functions. The check: `./segc sag/etale-flat all`, 121 modules, 6742 declarations, no failure, about 2 seconds.